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ACADEMIC VAULT

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ARC 101: Introduction to Architecture - Design Thinking, Drawing and Site

From the design process to scale drawings, orientation and the language of buildings

FACULTY PRIMER

Faculty Primer

Faculty of Environmental Sciences
Architecture | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Architecture freshers meet two demands at once: thinking conceptually and drawing precisely. This pack explains the design process, how to read and produce scale drawings, site and climate considerations, and the vocabulary you need to critique a building intelligently.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe the stages of the architectural design process from brief to detail.
- Read and produce measured drawings using scale, plans, sections and elevations.
- Analyse a site for orientation, drainage, access, noise and climate.
- Use architectural vocabulary accurately when presenting or critiquing work.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The design process

Design is not decoration applied at the end. It is a sequence of decisions from problem to built response.

- Brief: what the client needs, who will use the building, what constraints exist.
- Site analysis: orientation, access, topography, vegetation, noise, views and services.
- Concept: the organising idea - a diagram, a spatial strategy, a response to climate or culture.
- Schematic design: plans, sections and massing at small scale.
- Design development: materials, structure, services and detail coordination.
- Presentation: drawings, physical or digital models, and a clear verbal explanation.

Concept is not decoration
A concept explains why the building is organised the way it is. 'I used curves because they are beautiful' is a preference, not a concept.

Drawing and scale

Architectural drawing is a precise language. Learn the conventions and your work becomes legible to every reviewer.

- Plan: a horizontal cut about 1.2 metres above floor level.
- Section: a vertical cut showing heights and construction.
- Elevation: an external face of the building, without perspective.
- Scale: 1:100 or 1:50 for design drawings; 1:20 or 1:5 for details.
- Line weights: thin for construction lines, heavy for cut elements, medium for visible edges.
- Always include a north arrow, scale bar, title block and human figure for scale.

Site, climate and comfort

In a hot-humid climate such as Awka, passive design is not optional. Orientation and shading decide whether a room is usable.

- Long axis east-west reduces direct sun on long facades.
- Deep verandahs, brise-soleil and shaded openings reduce heat gain.
- Cross ventilation needs inlet and outlet openings on opposite sides.
- Consider prevailing winds, rainfall intensity and drainage before placing openings.
- Access and circulation: separate pedestrian and vehicle entry, and respect setbacks.
- Check the site's slope; it determines plinth levels and drainage strategy.

Critique language
Say 'the circulation is legible' or 'the service core is poorly located'. Avoid 'nice' and 'ugly'. Describe the effect, then judge it.

Architecture as response to conditions

A building is an answer to a set of conditions: climate, site, materials, budget, function, culture and regulation. Architectural quality is not decoration applied afterwards; it is how well the answer addresses those conditions at once.

In Nigeria the conditions are specific and demanding. Heat and humidity require shading, cross-ventilation and thermal mass; heavy rain requires overhangs and drainage; unreliable power requires daylighting strategies; and informal settlement requires incremental, affordable construction rather than imported models.

The classic triad - firmitas, utilitas, venustas (firmness, commodity, delight) - remains a useful test. A building that is beautiful and leaks has failed; so has one that is durable and uninhabitable.

Drawing, scale and representation

Architects think through drawings. Plans show a horizontal cut, sections a vertical cut, and elevations the external faces. Each is a convention, and each omits information the others carry; reading them together is how a building is understood.

Scale is the relationship between drawing and reality. A 1:100 plan means one unit on paper equals one hundred in reality. Getting scale wrong at design stage multiplies through construction, which is why dimensioning and checking are treated as basic professional habits.

Technical drawing conventions - line weights, hatching, dimension lines, north arrows - are not fussiness. They are the language in which a design is communicated to builders, engineers and clients.

> **Draw the sun path**
> Before placing a window, know where the sun is at 9 a.m. and 3 p.m. in the hot season. Orientation is free; air conditioning is not.

Structure, materials and sustainability

Structural systems - load-bearing walls, frames, trusses, shells - determine spans and therefore what spaces are possible. Understanding how loads travel from roof to foundation is what makes a design buildable, and it is the usual dividing line between a sketch and an architectural proposal.

Materials have properties that prescribe details: concrete is strong in compression, steel in tension; timber moves; blockwork needs movement joints; roofs need ventilation. Local materials - laterite, burnt brick, bamboo, thatch, stone - have environmental and economic advantages when detailed correctly.

Sustainability is now a design requirement, not an optional virtue. Passive strategies come first: orientation, shading, ventilation, thermal mass, daylight. Active systems and renewable energy come second, because they cost more to install and to maintain.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Designing for passive cooling

You must place a classroom block on a site in Enugu. Make the climate decisions.

1. Orient the long axis east-west so the long facades face north and south.
2. Shade the north and south facades with deep overhangs and horizontal fins.
3. Place openings to allow cross-ventilation from the prevailing breeze.
4. Use a high-level vent or clerestory to release hot air.
5. Choose a light-coloured roof with ventilation to reduce heat gain.
6. Place trees and covered walkways on the west side to block afternoon sun.

> **What to carry forward**
> Orientation, shading, ventilation, thermal mass. In that order, before any equipment.

2. Reading a set of drawings

You are given a plan, a section and an elevation. What do you check first?

1. Confirm the scale and the north point.
2. Check that the plan dimensions match the section heights and the storey count.
3. Trace circulation: can you reach every room without passing through another?
4. Check daylight and ventilation openings against room depth.
5. Identify structural elements and confirm they align across floors.
6. Note discrepancies and raise them before construction.

> **What to carry forward**
> Cross-check drawings against each other. Discrepancies found on paper are free to fix.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Explain how climate should shape building form in a hot-humid region.

20 marks

1. Describe the climate: high temperature, high humidity, heavy rain.
2. Orientation and shading strategy.
3. Ventilation: cross-ventilation and stack effect.
4. Roof design: overhangs, reflective surfaces, ventilation.
5. Material choices and thermal mass considerations.
6. Conclude that passive design precedes mechanical cooling.

Q2. Why are plans, sections and elevations all required to describe a building?

10 marks

1. Each shows one kind of information.
2. The plan gives layout; the section gives height and structure; the elevation gives appearance.
3. Cross-checking reveals errors.
4. Conventions exist so that others can read the design.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Plan. A drawing of a horizontal section showing layout at a stated scale.

Section. A drawing of a vertical cut showing heights and construction.

Elevation. A drawing of an external face of a building.

Scale. The ratio between a drawing and the real object.

Cross-ventilation. Air movement through openings on opposite or adjacent sides of a space.

Stack effect. Vertical air movement driven by temperature differences.

Thermal mass. A material's capacity to absorb and store heat, moderating temperature swings.

Load-bearing wall. A wall that carries structural loads from above.

Span. The distance a structural element bridges without intermediate support.

Firmitas, utilitas, venustas. Firmness, commodity and delight - the classical triad of architectural value.

Incremental housing. Housing designed to be extended and improved by residents over time.

Passive design. Using form, orientation and materials to manage climate without mechanical systems.

Your four-week plan

Week 1 - Conditions

Site and climate.

- Analyse one site: sun, wind, rain, access.
- Sketch a sun path.
- List five climate responses.

Week 2 - Drawing

Representation.

- Draw a plan, section and elevation at 1:100.
- Dimension correctly.
- Cross-check for consistency.

Week 3 - Structure

Buildability.

- Identify the structural system in a local building.
- Explain how loads reach the ground.
- Choose materials and justify.

Week 4 - Proposal

Integration.

- Design a small building to fit the brief.
- Show passive strategies on the drawings.
- Present orally in five minutes.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any introduction to architectural design covering drawing conventions and scale.
- A text on climate-responsive design in tropical regions.
- A structures primer for architects on load paths and spanning systems.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. What is the difference between a plan and an elevation?

Answer: A plan is a horizontal section cut through the building showing spaces and walls from above. An elevation is an orthographic view of an external face showing heights without perspective.

2. Why is a north arrow essential on a drawing?

Answer: It shows orientation, which determines solar exposure, prevailing wind and daylight strategy.

3. Give two passive strategies for a hot-humid climate.

Answer: Cross ventilation and shading with wide roof overhangs or brise-soleil. Others include light-coloured roofs and raised floors.

4. What should a site analysis cover?

Answer: Access, topography, orientation, climate, vegetation, drainage, noise, views, neighbouring buildings and available services.

5. What is a concept in architecture?

Answer: The organising idea that explains the design decisions and how the building responds to brief, site and use.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Measure your room and draw it to 1:50 with correct line weights.
- [ ] Draw the plan, section and one elevation of a simple building.
- [ ] Analyse your hostel site for sun, wind, access and drainage.
- [ ] Collect ten architectural terms and write a one-line definition for each.
- [ ] Sketch three concepts for a small campus kiosk with written justification.
- [ ] Visit one building on campus and critique its circulation in 150 words.
- [ ] Ask your studio tutor to review your drawing conventions before submission.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date: